

Signature in sound-mode of the exciton bilayer two-dimensional superfluid transition

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Obtaining definitive evidence of exciton superfluidity in electron–hole bilayers in zero magnetic field remains a major longstanding challenge since the condensate is electrically neutral, making its phase coherence difficult to detect directly. We show that the Anderson–Bogoliubov sound velocity provides a dynamical signature of exciton superfluidity. Across the BCS–BEC crossover, the velocity is known to discontinuously drop to zero at the Berezinskii–Kosterlitz–Thouless (BKT) transition. The magnitude of the drop has a strong density dependence. We compute this behavior, with the inclusion of finite-temperature screening, and determine the BKT transition using a renormalization-group approach. We further identify a temperature window which is experimentally accessible, where vortex–antivortex excitations strongly renormalize both the sound velocity and the transition temperature.

There are growing experimental indications of superfluidity or Bose–Einstein condensation (BEC) in semiconductor electron–hole double layer heterostructures in zero magnetic field [1–5]. Electron–hole bilayers are particularly interesting because exciton condensed phases may occur at relatively high temperatures [6], potentially leading to low-dissipation, electrically tunable devices [7–9]. Exciton bilayers also provide a versatile platform to explore the BCS–BEC crossover: by tuning the carrier density with external gates, the system can be driven from a strongly coupled BEC regime of tightly bound excitons to a weakly coupled BCS regime of overlapping electron–hole pairs [10–12].

The strong circumstantial indications of the presence of a superfluid or BEC, i.e. enhanced interlayer tunneling [1, 2, 13], near perfect Coulomb drag [5, 14–16], and, recently, spin-valley susceptibility in TMD bilayers [17], are consistent with exciton condensation. However, none of these directly establishes existence of a superfluid. Definitive confirmation is elusive and a great challenge, particularly since in a neutral exciton superfluid there is no Meissner effect.

We propose employing the Berezinskii–Kosterlitz–Thouless (BKT) transition [18, 19] to develop an experimentally accessible signature for exciton superfluidity. The BKT transition is a topological transition in which bound vortex–antivortex pairs unbind and destroy the quasi-long-range phase coherence in two-dimensional superfluids. Below the BKT transition temperature T_{BKT} , the finite phase stiffness supports a long-wavelength Anderson–Bogoliubov (AB) phase mode [20]. At T_{BKT} , the Nelson–Kosterlitz universal jump drives the renormalized stiffness to zero [21], and since the AB sound velocity is proportional to the square root of the stiffness, there is a discontinuous collapse of the AB velocity at the transition. The collapse of the AB mode can thus provide a fingerprint for the superfluid.

We have developed a finite-temperature path-integral theory for electron–hole bilayers. This combines temperature-dependent Random Phase Approximation (RPA) screening with a renormalization-group (RG) treatment of the vortex–antivortex fluctuations. With this we evaluate the BKT transition temperature, the renormalized stiffness, and the AB sound mode. In this way we have a route to identify in a neutral exciton condensate, both the superfluid transition and the vortex-fluctuation regime.

The electron–hole bilayer system consists of two n -doped and p -doped conducting parallel layers that are separated by a thin insulating layer (Fig. 1). A particle-hole transformation is used to map the empty electron states in the valence band of the p -doped layer to a conduction band populated by positively charged holes. The system is then modeled as two oppositely charged species interacting through Coulomb interactions.

The speed of sound is given by the derivative of the AB phase collective mode in the long-wave regime [22, 23], and so we start by generalizing the phase-only action for long-range Coulomb interactions. We set $k_B = \hbar = 1$ throughout this work. We focus on the low-density regime where the intralayer repulsion is negligible [24]. For a superfluid exciton bilayer, the finite-temperature path-integral action is $S = S_0 + S_{\text{int}}$ where

$$S_0 = \sum_{\lambda, \sigma} \int d\tau \int d\mathbf{x} \Psi_{\lambda, \sigma, \tau}^\dagger(\mathbf{x}) (\partial_\tau + T_\lambda - \mu_\lambda) \Psi_{\lambda, \sigma, \tau}(\mathbf{x}), \quad (1)$$

$$S_{\text{int}} = \int d\tau \int d\mathbf{x} \int d\mathbf{y} \left[\sum_{\sigma \neq \sigma'} \left(\Delta_\tau^*(\mathbf{x}, \mathbf{y}) \Psi_{e, \sigma, \tau}(\mathbf{x}) \Psi_{h, \sigma', \tau}(\mathbf{y}) + \Delta_\tau(\mathbf{x}, \mathbf{y}) \Psi_{e, \sigma, \tau}^\dagger(\mathbf{x}) \Psi_{h, \sigma', \tau}^\dagger(\mathbf{y}) \right) + \frac{|\Delta_\tau(\mathbf{x}, \mathbf{y})|^2}{V_{eh}(\mathbf{x}, \mathbf{y})} \right]. \quad (2)$$

S_0 is the non-interacting term and S_{int} the attractive interaction term in the pairing (Bogoliubov) channel [25]. The

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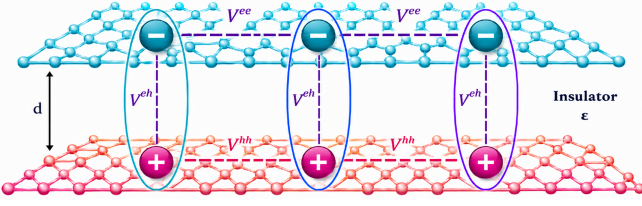


Figure 1. Schematic of the exciton bilayer system. Top layer of electrons, bottom layer of holes. Separated by an insulator of dielectric constant ϵ and thickness d .

fermionic fields $\Psi_{\lambda,\sigma,\tau}(\mathbf{x})$ are functions of the inlayer position $\mathbf{x}$ and the imaginary time τ , and are labeled by the spin-valley index $\sigma = (v, s_z)$ and $\lambda = e, h$ for the electron and hole layers. In Eq. (1) the kinetic operator $T_\lambda = -\nabla^2/(2m^*)$ and the chemical potentials are μ_λ . We consider only equal electron and hole effective masses $m^* = m_e^* = m_h^*$ and equal densities $n = n_e = n_h$. In Eq. (2), the bare inter-layer Coulomb attraction is $V_{eh}(\mathbf{x}, \mathbf{y}) = -e^2/(4\pi\epsilon)(d^2 + |\mathbf{x} - \mathbf{y}|^2)^{-1/2}$, with d the layer separation and ϵ the permittivity of the insulating spacer layer [26]. Equation (2) is expressed in terms of the Hubbard-Stratonovic pair-field $\Delta_\tau(\mathbf{x}, \mathbf{y}) = V_{eh}(\mathbf{x}, \mathbf{y})\langle\Psi_{e,\sigma,\tau}(\mathbf{x})\Psi_{h,\sigma',\tau}(\mathbf{y})\rangle$. Since the inter-layer Coulomb interaction is nearly spin independent, singlet and triplet excitonic channels are considered degenerate [27]. The interaction is decoupled in the spin-valley singlet inter-layer exciton channel ($\sigma \neq \sigma'$) [6, 28, 29], allowing us to use a spin-valley scalar order parameter rather than the full spin matrix $\Delta_{\sigma\sigma'}$. The singlet projection is a simplifying choice and not imposed by any material selection rule.

The phase dependence in S is made explicit by performing a spin-independent gauge transformation on the fermionic fields, $\Psi_{\lambda,\sigma,\tau} \rightarrow \Psi_{\lambda,\sigma,\tau} e^{i\theta(\mathbf{x},\tau)/2}$. The non-interaction part of the action, Eq. (1), transforms as $S_0 \rightarrow S_0 + S_{\text{ph}}$ where,

$$S_{\text{ph}}[\theta] = \sum_{\lambda,\sigma} \int_0^\beta d\tau \int d\mathbf{x} \Psi_{\lambda,\sigma,\tau}^\dagger(\mathbf{x}) \times \left(\frac{i}{2} \partial_\tau \theta + \frac{1}{8m^*} (\nabla \theta)^2 + \frac{i}{4m^*} \nabla \theta \cdot \overleftrightarrow{\nabla} \right) \Psi_{\lambda,\sigma,\tau}(\mathbf{x}), \quad (3)$$

with $\nabla \theta \cdot \overleftrightarrow{\nabla} \Psi_{\lambda,\sigma,\tau}(\mathbf{x}) = \nabla^2 \theta \cdot \Psi_{\lambda,\sigma,\tau}(\mathbf{x}) + \nabla \theta \cdot \nabla \Psi_{\lambda,\sigma,\tau}(\mathbf{x})$. The interaction part of the action, Eq. (2), is left invariant since the pairing field transforms under the gauge transformation as $\Delta_\tau(\mathbf{x}, \mathbf{y}) = |\Delta_\tau(\mathbf{x}, \mathbf{y})| e^{i[\theta(\mathbf{x},\tau) + \theta(\mathbf{y},\tau)]/2}$. Thus, for a Coulomb interaction, as for a contact interaction, the phase field θ enters only through the noninteracting term S_{ph} . This occurs for any interaction that is independent of spin and independent of the pair center of mass coordinate, and in the absence of spin-orbit coupling [30]. The present derivation extends the result for a separable interaction [31] to a nonseparable long-range interaction.

We now separate the action into mean-field and phase-fluctuation contributions, expanding around the superfluid saddle-point [31–34]. In the hydrodynamic limit correspond-

ing to the long-wavelength and low-frequency regime, $q\xi \ll 1$ and $\omega \ll 2\Delta$ with ξ the healing length, the Anderson–Bogoliubov branch is acoustic at small momentum, $\omega(q) = c_s q$ [31]. The sound velocity c_s is temperature dependent and is given by,

$$c_s(T) = \sqrt{\frac{J_0(T)}{\kappa_0(T)}}, \quad (4)$$

where $\kappa_0 = \kappa/2$ and $\kappa = \frac{\partial n}{\partial \mu}|_{T,V}$ is the compressibility [31–33]. The phase stiffness is,

$$J_0(T) = \frac{n_X}{m_X^*} - \frac{g}{A} \sum_{\mathbf{k}} \left(\frac{k_x^2}{m_X^*} \right) n_F(E_{\mathbf{k}}, T). \quad (5)$$

$n_F(E_{\mathbf{k}}, T) = 1/(e^{E_{\mathbf{k}}(T)/T} + 1)$ is the Fermi-Dirac distribution function, and $E_{\mathbf{k}}(T) = \sqrt{\epsilon_{\mathbf{k}}^2 + |\Delta_0(\mathbf{k}, T)|^2}$ is the superfluid excitation energy with $\epsilon_{\mathbf{k}} = \mathbf{k}^2/2m^* - \mu$. A is the area of each layer, k_x is the x -component of the vector momentum $\mathbf{k}$. $n_X = n$ and $m_X^* = 2m^*$ are the exciton density and effective mass. Since the Coulomb interaction is spin and valley independent, the σ index can be omitted, with the spin and valley degeneracy accounted for by a prefactor g .

At zero temperature, Eq. (5) reduces to $J_0(0) = n/(2m^*)$. This is twice the BCS value for an electron gas with electron density n since in a BCS superconductor two electrons form a single Cooper pair, whereas in an electron–hole bilayer each electron binds one hole to form one exciton, so $n_X = n$. This one-to-one conversion of carriers into excitons gives the exciton condensate a bare stiffness that is a factor of two larger than for the BCS Cooper-pair condensate. Since T_{BKT} is proportional to the stiffness, this provides a significant intrinsic advantage for exciton bilayers, allowing substantially higher BKT transition temperatures. Similarly, the compressibility of the excitons is twice the BCS compressibility.

The stiffness and compressibility depend on the saddle-point values Δ_0 and μ . These are obtained from the gap and number equations that follow from minimizing the mean-field contribution S_{MF} of the action at temperature T ,

$$\Delta_0(\mathbf{k}, T) = -\frac{1}{A} \sum_{\mathbf{k}'} V_{eh}^{RPA}(\mathbf{k} - \mathbf{k}', T) \frac{\Delta_0(\mathbf{k}', T) \tanh\left(\frac{E_{\mathbf{k}'}(T)}{2T}\right)}{2E_{\mathbf{k}'}(T)} \quad (6)$$

$$n_{MF} = \frac{g}{A} \sum_{\mathbf{k}} \frac{1}{2} \left(1 - \frac{\epsilon_{\mathbf{k}} \tanh\left(\frac{E_{\mathbf{k}}(T)}{2T}\right)}{E_{\mathbf{k}}(T)} \right). \quad (7)$$

V_{eh}^{RPA} is the static screening within RPA [35] (see Appendix). We now include phase fluctuations by decomposing the phase field into $\theta = \theta_{\text{sw}} + \theta_v$, where θ_{sw} denotes smooth spin-wave fluctuations and θ_v is the vortex contribution [33]. For the range of densities considered here, T_{BKT} lies well below the temperature scale at which spin-waves appreciably affect the density equation so their contribution can be neglected (for

details see the Supplemental material). The transition is therefore controlled by the vortex–antivortex fluctuations.

For $T < T_{\text{BKT}}$, bound vortex–antivortex pairs renormalize the phase stiffness as the temperature increases. At $T = T_{\text{BKT}}$, the pairs unbind, destroying the quasi-long-range order and producing the universal stiffness jump [21]. As already noted, this is directly reflected in the AB sound velocity $c_s(T)$ (Eq. (4)). A first estimate of the BKT transition temperature is given by the Nelson–Kosterlitz criterion $T_{\text{BKT}}^0 = \frac{\pi}{2} J_0(T_{\text{BKT}}^0)$ [21].

To account for the vortex-induced renormalization of the stiffness, we solve the Kosterlitz–Thouless RG equations [21]: $\partial_\ell K_\ell^{-1} = 4\pi^3 y_\ell^2$ and $\partial_\ell y_\ell = (2 - \pi K_\ell) y_\ell$. $K_\ell(T) = J_\ell(T)/T$ and $y_\ell(T) = \exp(-\mu_{\text{vor},\ell}/T)$ is the vortex fugacity. $J_{\ell=0}(T) \equiv J_0(T)$ (Eq. (5)).

The bare vortex-core energy $\mu_{\text{vor},\ell}$ is determined microscopically from the solution of the Gross–Pitaevskii (GP) equation for a dilute two-dimensional exciton gas. We solve the GP equation using the nonlocal exciton–exciton interaction and obtain the corresponding vortex profile and vortex core energy. In the deep BEC regime and for the range of densities considered here (details in the Appendix), we find no appreciable deviation from the standard estimate $\mu_{\text{vor},0}(T) = 2.45 J_0(T)$, obtained for a 2D Bose gas with purely contact interactions [36]. We therefore use these values for $\mu_{\text{vor},0}(T)$. The RG-renormalized transition temperature is then,

$$T_{\text{BKT}}^{\text{RG}} = \frac{\pi}{2} J_{\text{RG}}(T_{\text{BKT}}^{\text{RG}}), \quad (8)$$

with $J_{\text{RG}}(T) \equiv J_{\ell \rightarrow \infty}(T)$ the fully renormalized stiffness.

We now investigate for the first time the properties of the exciton superfluid at finite temperatures. First the gap and number equations are solved in the low-density BEC regime as a function of temperature and then the temperature dependence of the superfluid RPA screening is determined.

Figure 2(a) shows the temperature dependence of the maximum gap Δ_{max} for double-bilayer graphene (DBG). We take $m_* = 0.045 m_e$, $g = 4$, $\varepsilon = 3\varepsilon_0$ [37, 38]. The effective Bohr radius is $a_B^* = 7.9 \text{ nm}$ and the effective Rydberg $Ry^* = 30 \text{ meV}$. The interlayer separation is fixed at $d = 2 \text{ nm}$ and the densities n are labeled by the average inlayer interparticle distances $r_0 = 1/\sqrt{\pi n}$. We restrict the $r_0 > 1.78 a_B^*$, for which the zero-temperature Δ_{max} has not yet reached its maximum as a function of density, so Hartree–Fock intralayer contributions can be neglected [24].

At low density, Δ_{max} decreases smoothly with temperature before collapsing at the mean-field critical temperature T_c^{MF} . Unlike in conventional BCS superconductors, the exciton superfluid mean-field transition is not driven primarily by thermal pair breaking but by the weakening of the electron–hole attraction due to increased screening. With increasing density, the precursor decrease is progressively lost and the collapse due to superfluid screening becomes sharper. This behavior is controlled by the temperature-dependent superfluid screening. In Fig. 2(b), this is shown through the cancellation between the normal and anomalous static polarizations,

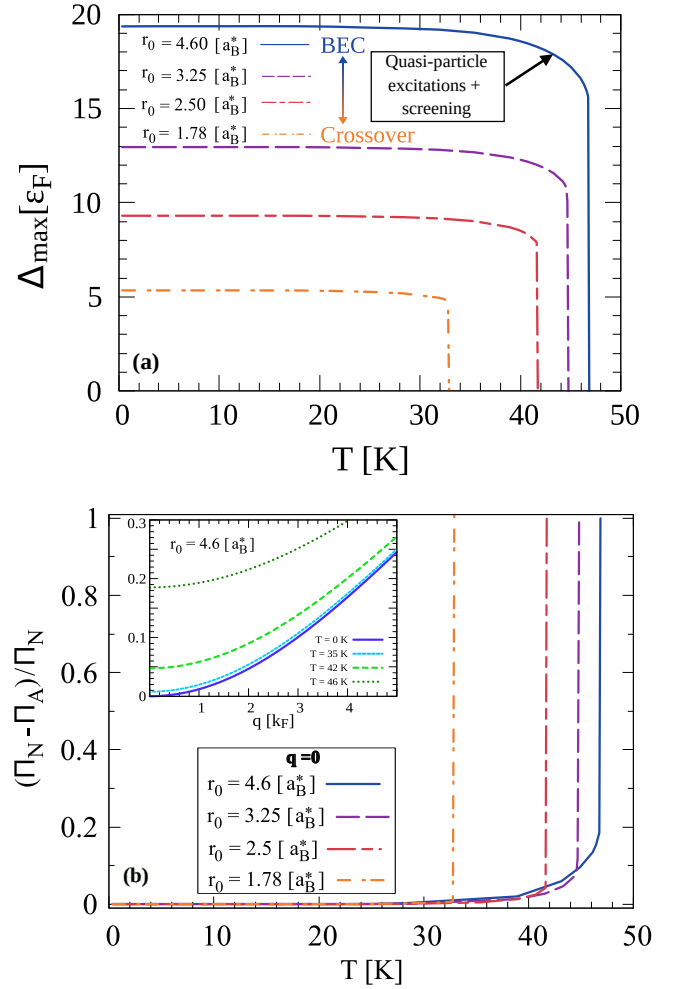


Figure 2. (a) Maximum gap energy Δ_{max} as a function of the temperature T . The r_0 are the inlayer interparticle distances for different densities. (b) Difference between the normal and anomalous polarization functions $[\Pi_N - \Pi_A]$, normalized to Π_N , $\mathbf{q} = 0$ as a function of T . Inset: $[\Pi_N - \Pi_A]/\Pi_N$ for $r_0 = 4.6 a_B^*$ as a function of $\mathbf{q}$ for different T .

$\Pi_N(\mathbf{q}) - \Pi_A(\mathbf{q})$. The more complete the cancellation for $q \lesssim 2k_F$, the weaker the screening [35].

At $T = 0$, the cancellation is exact at $\mathbf{q} = 0$ [39] and, at low density, remains nearly complete up to $q \simeq 2k_F$ (inset of Fig. 2(b)), strongly suppressing screening. At finite temperature, thermal pair breaking introduces excited electrons and holes through the pair-breaking terms in Eqs. (11)–(12), so the cancellation is no longer exact even at $\mathbf{q} = 0$. For $r_0 = 4.6 a_B^*$, the cancellation remains strong up to $T \simeq 35 \text{ K}$ and then rapidly weakens as thermally excited carriers enhance screening. This drives the suppression and eventual collapse of Δ_{max} at T_c^{MF} .

The density dependence of T_c^{MF} follows from the ratio $\Delta_{\text{max}}/\varepsilon_F$. In the dilute regime this ratio is large, so thermal quasiparticle excitation is costly and screening turns on only gradually. At higher density, $\Delta_{\text{max}}/\varepsilon_F$ is smaller and more carriers contribute to screening; once the $\mathbf{q} = 0$ cancellation

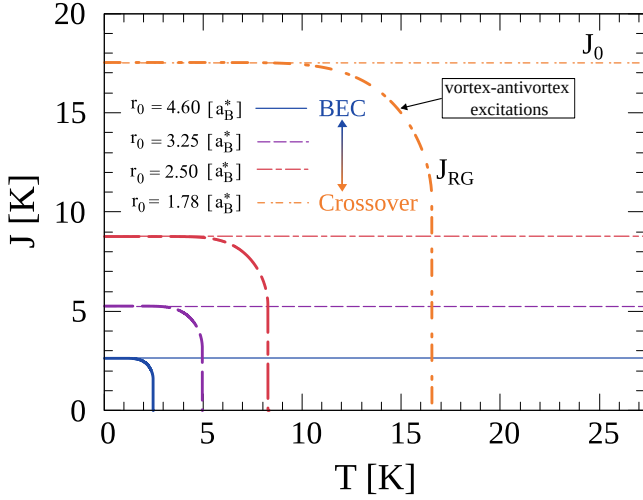


Figure 3. Stiffness $J_0(T)$ (horizontal lines) and renormalized stiffness $J_{RG}(T)$ as a function of temperature T for interparticle distances r_0 , as labeled.

is lost, screening is rapidly enhanced over $0 < q < 2k_F$, producing an abrupt gap collapse.

In Fig. 3 we compare the bare stiffness $J_0(T)$ (Eq. (5)), with the renormalized stiffness $J_{RG}(T)$. At low temperatures, $J_0(T)$ and $J_{RG}(T)$ coincide since there are a negligible number of thermally excited vortex–antivortex pairs. $J_0(T)$ remains practically temperature independent for all $T < T_c^{MF}$, reflecting the negligible quasiparticle contribution to Eq. (5), leaving $J_0 \simeq n/(2m^*)$. At $T = T_c^{MF}$, $J_0(T)$ then drops suddenly to zero. In contrast, $J_{RG}(T)$ is progressively suppressed with increasing temperature due to the proliferation of thermally excited vortex–antivortex pairs in the superfluid. At the renormalized BKT transition temperature T_{BKT}^{RG} , vortex–antivortex unbinding occurs, the stiffness collapses to zero, and the coherent superfluid phase is destroyed. We can conclude that in this system the vortex–antivortex interaction and the quasi-particle contribution are disentangled in the superfluid.

A central result in Fig. 3 is that in exciton bilayers, the vortex-renormalization temperature window for $J_{RG}(T)$ extends over a number of degrees K for both the BEC and BEC-BCS crossover regimes. This is a striking result for a solid-state platform, recalling that in weak-coupled BCS superconductors, the vortex–antivortex fluctuations are usually confined to a mK scale close to the transition [40, 41], making the fluctuations very difficult indeed to isolate experimentally.

Figure 4 shows the temperature–density phase diagram for DBG. For $T < T_{BKT}^{RG}$, the system is an exciton superfluid, while for $T > T_c^{MF}$ it is a normal electron–hole liquid. In the intermediate temperature interval, $T_{BKT}^{RG} < T < T_c^{MF}$, there is still local pairing but unbinding of the vortex–antivortex pairs has destroyed the superfluid phase [42–44]. We denote this regime in Fig. 4 as the pseudogap. The temperature–density window over which vortex excitations renormalize the stiffness is also indicated. T_{BKT}^0 is the unrenormalized transi-

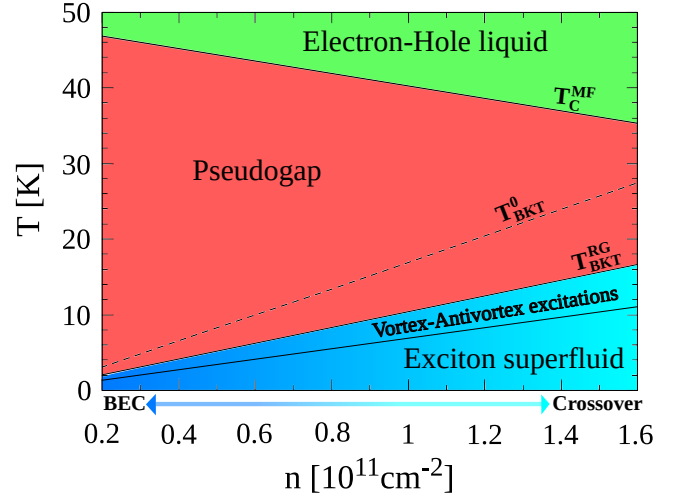


Figure 4. Temperature–density (T, n) phase-diagram for double-bilayer-graphene. T_c^{MF} : mean-field critical temperature. T_{BKT}^0 and T_{BKT}^{RG} : BKT transition temperature without and with RG.

tion temperature (Eq. (8)).

The RG flow significantly suppresses T_{BKT}^{RG} , typically reducing it by a factor of two relative to T_{BKT}^0 . Thus vortex renormalization is a leading and highly significant correction. T_{BKT}^{RG} increases approximately linearly with density, since over the density range considered $T_{BKT}^{RG} \ll T_c^{MF}$. Hence quasiparticle excitations are exponentially suppressed, leaving vortex–antivortex excitations as the dominant mechanism in renormalizing the stiffness.

The compressibility κ for $T < T_{BKT}^{RG}$ is likewise practically temperature independent, so from Eq. (4), the AB sound velocity $c_s(T)$ rigidly follows $\sqrt{J_{RG}(T)}$, rendering vortex–antivortex fluctuations directly visible from the sound velocity in Fig. 5.

When the temperature is increased, c_s becomes steadily more suppressed by thermally excited vortex pairs. It drops discontinuously to zero at T_{BKT}^{RG} . With increasing density, $c_s(T)$ and T_{BKT}^{RG} both increase as they track $J_{RG}(T)$ (Fig. 3). Also shown in Fig. 5, is the zero-temperature sound velocity for composite-bosons, $\bar{c}_s = \sqrt{\frac{2\mu_s + \varepsilon_B/2}{m^*}}$ [45] at $r_0 = 4.6 a_B^*$, with ε_B our calculated exciton binding energy. We note the excellent agreement with our calculated $c_s(T = 0)$. This confirms the bosonic character of the exciton low-density regime.

The drop ($\sim$ Fermi velocity) of c_s near T_{BKT}^{RG} occurs over several Kelvin and makes the vortex-renormalization regime experimentally resolvable. Since κ is almost temperature independent below T_{BKT}^{RG} , measuring $c_s^2(T)$ directly tracks $J_{RG}(T)$, providing access to the renormalized stiffness. The collapse of c_s is large and is our proposed signature of both the BKT transition and the existence of the exciton superfluid.

The mode could be probed through density-response techniques, including Bragg spectroscopy [46, 47] or through phase-sensitive schemes based on weak coupling to a coherent reservoir [48, 49].

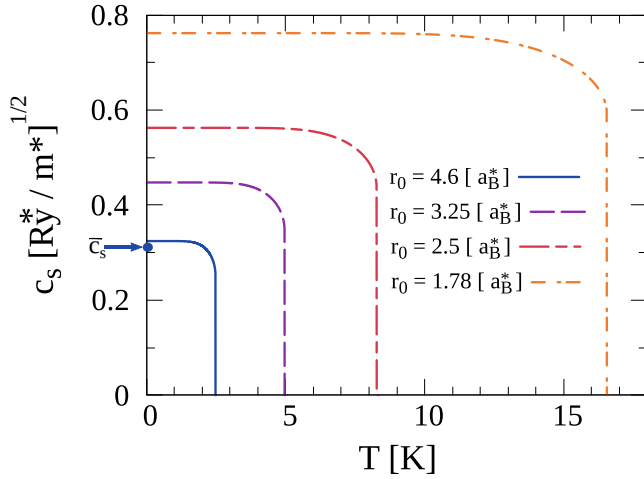


Figure 5. Speed of sound c_s for AB mode as a function of temperature T for different interparticle distances r_0 , as labeled. $\bar{c}_s = \sqrt{2\mu_s + \varepsilon_B/2m^*}$ for $r_0 = 4.6a_B^*$, obtained from composite boson theory at $T = 0$.

To summarize, the temperature-dependent screening in the superfluid arising from thermal pair-breaking, drives the mean-field transition to the normal electron-hole liquid. Our temperature–density exciton phase diagram includes a RG vortex–antivortex excitation region which joins the superfluid, pseudogap, and normal electron–hole liquid phases [50]. Remarkably, the vortex-antivortex region extends over several degrees Kelvin, in contrast to the few-mK scale typical of solid-state superconductors [40, 41, 51]. This region, where deviations from mean-field caused by vortex–antivortex pairs in the RG flow, is thus experimentally accessible in exciton bilayers. The AB sound velocity $c_s(T)$ is steadily suppressed with increasing temperature by the vortex–antivortex excitations. This is followed by a sudden drop to zero at the BKT transition temperature. We demonstrate that the temperature-dependence of the superfluid stiffness, for which magnetic penetration depth measurements are unavailable, is directly obtainable from $c_s(T)$. From this, one can obtain the temperature window where vortex-antivortex RG excitations are significant.

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End Matter

Appendix 1: Finite temperature RPA screened interaction

We include the superfluid screening in the gap equation, Eq. (6), by evaluating the screened attractive electron-hole interaction V_{eh}^{RPA} within the self-consistent finite-temperature RPA [35],

$$V_{eh}^{RPA}(\mathbf{q}, T) = \frac{V_{\mathbf{q}}^D + \Pi_{\mathbf{q}}^A(T) \mathcal{A}_{\mathbf{q}}}{1 - 2(\Pi_{\mathbf{q}}^N(T) V_{\mathbf{q}}^S + \Pi_{\mathbf{q}}^A(T) V_{\mathbf{q}}^D) + \mathcal{B}_{\mathbf{q}}(T) \mathcal{A}_{\mathbf{q}}} \quad (9)$$

with

$$\mathcal{A}_{\mathbf{q}} = (V_{\mathbf{q}}^S)^2 - (V_{\mathbf{q}}^D)^2, \quad \mathcal{B}_{\mathbf{q}}(T) = (\Pi_{\mathbf{q}}^N(T))^2 - (\Pi_{\mathbf{q}}^A(T))^2. \quad (10)$$

$\Pi_{\mathbf{q}}^N(T)$ and $\Pi_{\mathbf{q}}^A(T)$ are the static normal and anomalous polarization functions at temperature T ,

$$\begin{aligned} \Pi_{\mathbf{q}}^N(T) = & \sum_{\mathbf{k}} (u_{\mathbf{k}}^2 u_{\mathbf{k}+\mathbf{q}}^2 + v_{\mathbf{q}}^2 v_{\mathbf{k}+\mathbf{q}}^2) \frac{n_F(E_{\mathbf{k}}, T) - n_F(E_{\mathbf{k}+\mathbf{q}}, T)}{E_{\mathbf{k}} - E_{\mathbf{k}+\mathbf{q}}} \\ & - (v_{\mathbf{k}}^2 u_{\mathbf{k}+\mathbf{q}}^2 + u_{\mathbf{k}}^2 v_{\mathbf{k}+\mathbf{q}}^2) \frac{1 - n_F(E_{\mathbf{k}}, T) - n_F(E_{\mathbf{k}+\mathbf{q}}, T)}{E_{\mathbf{k}} + E_{\mathbf{k}+\mathbf{q}}} \end{aligned} \quad (11)$$

$$\begin{aligned} \Pi_{\mathbf{q}}^A(T) = & -2u_{\mathbf{k}} v_{\mathbf{k}} u_{\mathbf{k}+\mathbf{q}} v_{\mathbf{k}+\mathbf{q}} \left[\frac{n_F(E_{\mathbf{k}}, T) - n_F(E_{\mathbf{k}+\mathbf{q}}, T)}{E_{\mathbf{k}} - E_{\mathbf{k}+\mathbf{q}}} \right. \\ & \left. + \frac{1 - n_F(E_{\mathbf{k}}, T) - n_F(E_{\mathbf{k}+\mathbf{q}}, T)}{E_{\mathbf{k}} + E_{\mathbf{k}+\mathbf{q}}} \right], \end{aligned} \quad (12)$$

where $v_{\mathbf{k}}^2 = 1/2(1 - \epsilon_{\mathbf{k}}/E_{\mathbf{k}})$ and $u_{\mathbf{k}}^2 = 1/2(1 + \epsilon_{\mathbf{k}}/E_{\mathbf{k}})$ are the Bogoliubov amplitudes. We use the abbreviated notation, $E_{\mathbf{k}}(T) \rightarrow E_{\mathbf{k}}$. The bare Coulomb interactions in the effective mass approximation for 2D bilayers of electrons (e) and holes (h), $V_{\mathbf{q}}^S = V_{\mathbf{q}}^{ee} = V_{\mathbf{q}}^{hh} = \frac{2\pi e^2}{4\pi\epsilon|\mathbf{q}|}$ and $V_{\mathbf{q}}^D = V_{\mathbf{q}}^{eh} = V_{\mathbf{q}}^{he} = -V_{\mathbf{q}}^S e^{-|\mathbf{q}|d}$ [26]. In Eqs. (11)-(12), the first term comes from pair-breaking due to thermal excitations. It vanishes at $T = 0$.

Appendix 2: Exciton bilayer vortex-core energy

The vortex energy is the energy systems difference $\Delta E \equiv E[\psi_v] - E[\psi_0]$, where $\psi_0 = \sqrt{n_0}$ is the uniform wave-function of the superfluid at fixed particle number density n_0 and ψ_v is the wave-function with one vortex. We recall for a contact interaction of coupling strength g , the Gross-Pitaevskij (GP) energy functional $E[\psi]$ is,

$$E[\psi] = \int d^2\mathbf{r} \left[\frac{1}{2m_X^*} |\nabla\psi|^2 + \frac{g}{2} |\psi|^4 \right]. \quad (13)$$

For ψ_0 , the kinetic term vanishes and $E[\psi_0] = \frac{g}{2} n_0^2 A$, with A the system area. For ψ_v , the linear term in $(|\psi_v|^2 - n_0)$ cancels from particle-number conservation. So,

$$\Delta E = \int d^2\mathbf{r} \left[\frac{1}{2m_X^*} |\nabla\psi_v|^2 + \frac{g}{2} (|\psi_v|^2 - n_0)^2 \right]. \quad (14)$$

For a singly quantized vortex centered at the origin, the ansatz

$$\psi_v(\mathbf{r}) = \sqrt{n_0} f(r/\xi) e^{i\vartheta}, \quad \xi^2 \equiv \frac{1}{2m_X^* g n_0} \quad (15)$$

is adopted, where ϑ is the azimuthal angle, $f(0) = 0$ and $f(x) \rightarrow 1$ as $x \rightarrow \infty$. Inserting Eq. (15) into Eq. (14), and performing the angular integration with a variable change $x = r/\xi$, the vortex-core energy E_c is obtained by discarding the centrifugal term f^2/x^2 ,

$$E_c = \frac{1}{2m_X^*} n_0 \pi \int_0^\infty dx x \left[2 \left(\frac{df}{dx} \right)^2 + (1 - f^2)^2 \right]. \quad (16)$$

We have extended the formalism to the exciton bilayer with non-contact interactions. The exciton GP energy functional is,

$$\begin{aligned} E_X[\psi] = & \int d^2\mathbf{r} \frac{1}{2m_X^*} |\nabla\psi|^2 \\ & + \frac{1}{2} \int d^2\mathbf{r} d^2\mathbf{r}' |\psi(\mathbf{r})|^2 V_{XX}(|\mathbf{r} - \mathbf{r}'|) |\psi(\mathbf{r}')|^2. \end{aligned} \quad (17)$$

where $V_{XX}(\mathbf{r})$ is the exciton-exciton interaction [52],

$$V_{XX}(\mathbf{r}) = \frac{2e^2}{4\pi\epsilon} \left(\frac{1}{\mathbf{r}} - \frac{1}{\sqrt{\mathbf{r}^2 + d^2}} \right). \quad (18)$$

$\mathbf{r}$ is the in-plane interparticle vector and d the layer separation. For $d \ll r_0$, the low density regime, V_{XX} is a dipolar interaction. The dipole moment is tunable through layer separation d .

The kinetic term in Eq. (17) is the same as for the contact interaction, but the interaction is a nonlocal double integral with kernel V_{XX} . We compared the vortex profile and core energy for contact and exciton-exciton interactions. For the contact interaction we chose an effective contact coupling g_{eff} defined as the zero-momentum limit of the dipole-dipole interaction,

$$V_{\text{dd}}(k) = 8\pi \frac{1 - e^{-dk}}{k}, \quad (19)$$

$$g_{\text{eff}} = \lim_{k \rightarrow 0} V_{\text{dd}}(k) = 8\pi d. \quad (20)$$

The units are $[\text{Ry}^* a_B^{*2}]$. This generates the contact interaction that reproduces the dipolar kernel in the long-wavelength limit.

For the DBG parameters used in the main text, and with $d = 2 \text{ nm} \simeq 0.253 a_B^*$, this gives $g_{\text{eff}} \simeq 6.36 \text{ Ry}^* (a_B^*)^2$. We use this value of g_{eff} to evaluate the local functional (Eq. (13)) and V_{XX} (Eq. (18)) to evaluate the nonlocal functional, Eq. (17). For the wave functions in Eqs. (13) and (17), we solved the GP equations for the contact interaction,

$$-\frac{1}{2m_X^*} \nabla^2 \Psi(\mathbf{r}) + g_{\text{eff}} |\Psi(\mathbf{r})|^2 \Psi(\mathbf{r}) = \mu \Psi(\mathbf{r}), \quad (21)$$

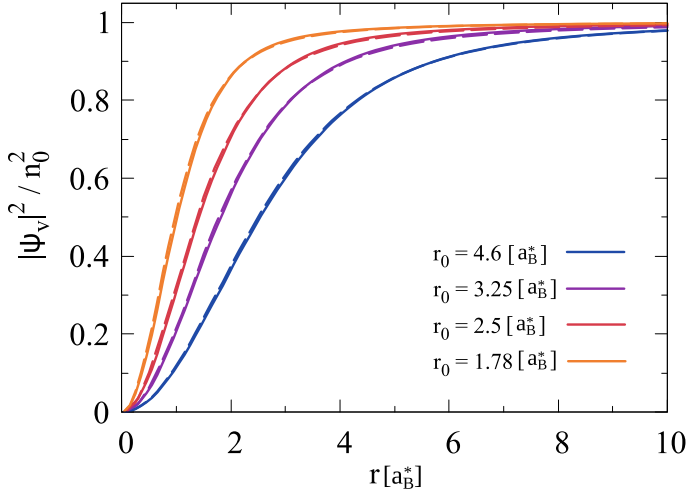


Figure 6. Vortex density profiles $|\psi_v(r)|^2/n_0^2$ as a function of the radial distance r , obtained from the imaginary-time solutions of the GP equations for contact interactions (solid lines) and dipolar interactions (dashed lines). The values of the inlayer interparticle distance r_0 span the density range addressed in this work.

and for the dipolar interaction,

$$-\frac{1}{2m_X^*}\nabla^2\Psi(\mathbf{r}) + \phi_{XX}(\mathbf{r})\Psi(\mathbf{r}) = \mu\Psi(\mathbf{r}), \quad (22)$$

where ϕ_{XX} is the exciton mean-field potential [52],

$$\phi_{XX}(\mathbf{r}) = \int V_{XX}(|\mathbf{r} - \mathbf{r}'|) |\Psi(\mathbf{r}')|^2 d\mathbf{r}'. \quad (23)$$

We solved Eqs. (21) and (22) by imaginary-time evolution [52, 53], starting from a randomized order parameter, and evaluating ϕ_{XX} in Fourier space through the convolution theorem [54]. The simulations were performed on a 2000^2 grid with Dirichlet boundary conditions. Convergence was reached when the relative energy change between successive iterations fell below 10^{-10} .

The vortex density profiles obtained with the contact and dipolar interactions are shown in Fig. 6 for four densities. In the low density regime which is our primary interest here, the two profiles are essentially indistinguishable. This reflects the fact that the excitons are compact and widely spaced which makes the vortex structure completely insensitive to whether the interaction is local or long-ranged.

For the contact interaction, the vortex-core energy is universal and density independent, with $E_c/J_0 \simeq 2.45$ [33, 55]. For the dipolar gas, we find a comparable value that remains almost constant over the density range studied. This is consistent with the vortex profiles in Fig. 6. Deviations from the contact-interaction result can be expected at higher densities or at larger interlayer separations, where the details of the non-local form of the dipolar interaction become important. In this regime, a peak can develop near the vortex edge, signaling an accumulation of superfluid density around the core [52].

Supplemental material: Spin-wave fluctuations

We expand the action $S = S_0 + S_{\text{int}}(|\Delta|) + S_{\text{ph}}$ around its saddle point by writing $|\Delta| = |\Delta_0| + \delta|\Delta|$ and neglecting amplitude fluctuations. The action then decomposes as,

$$S = S_{MF}(|\Delta_0|) + S_{\text{ph}}(\theta), \quad (24)$$

where $S_{MF} = S_0 + S_{\text{int}}(|\Delta_0|)$ is the mean-field action. In momentum space this reads,

$$S_{MF} = \sum_{\mathbf{k}, n} \ln[\det(-\mathbb{G}^{-1})_{\mathbf{k}, n}] + \beta \frac{1}{A} \sum_{\mathbf{k}, \mathbf{p}} \frac{|\Delta_0(\mathbf{k}, T)| |\Delta_0(\mathbf{p}, T)|}{V_{eh}(\mathbf{k} - \mathbf{p})}, \quad (25)$$

with $\mathbb{G}$ the Gor'kov single-particle propagator (details of the calculation are in Ref. [56]). The phase action takes the form,

$$S_{\text{ph}} = \frac{1}{2} \int_0^\beta d\tau \int d^2\mathbf{x} \left[J_0(T) (\nabla\theta)^2 + \kappa_0(T) (\partial_\tau\theta)^2 \right], \quad (26)$$

with $J_0(T)$ and $\kappa_0(T)$ defined in the main text.

The phase field θ can be decomposed into a longitudinal spin-wave component θ_{SW} and a transverse static vortex component θ_V ,

$$\theta(\mathbf{x}, \tau) = \theta_{SW}(\mathbf{x}, \tau) + \theta_V(\mathbf{x}). \quad (27)$$

θ_{SW} describes smooth phase fluctuations satisfying $\nabla^2\theta_{SW} = 0$, while θ_V encodes topological vortex configurations with quantized winding number, $\nabla^2\theta_V = \sum_i 2\pi m_i \delta^{(2)}(\mathbf{x} - \mathbf{x}_i)$, $m_i \in \mathbb{Z}$. The two contributions are orthogonal, so the phase action splits into $S_{\text{ph}} = S_{SW} + S_V$, with spin-wave contribution,

$$S_{SW} = \frac{1}{2} \int_0^\beta d\tau \int d^2\mathbf{x} \left[J_0(T) (\nabla\theta_{SW})^2 + \kappa_0(T) (\partial_\tau\theta_{SW})^2 \right], \quad (28)$$

and static vortex contribution

$$S_V = \frac{1}{2T} \int d^2\mathbf{x} J_0(T) (\nabla\theta_V)^2. \quad (29)$$

Equation (29) is formally equivalent to the classical XY action governing the BKT transition [31]. The vortex contribution S_V is treated within the renormalization-group approach used in the main text to determine the renormalized stiffness $J_{RG}(T)$ and the BKT transition temperature $T_{\text{BKT}}^{\text{RG}}$. It is not discussed further here. The gap equation then follows from minimizing the

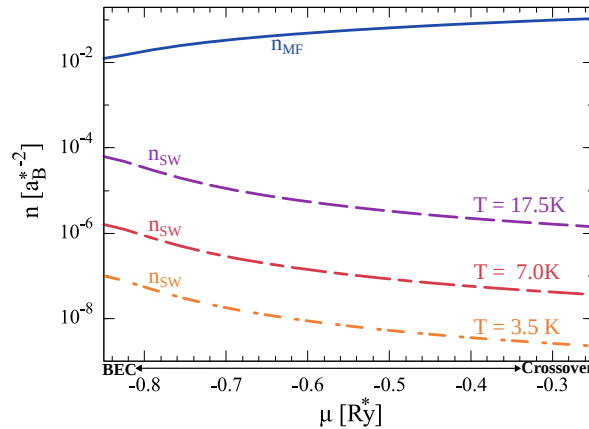


Figure 7. Mean-field contribution n_{MF} (solid line) and spin-wave contribution n_{SW} (dashed lines) to the density equation as a function of the chemical potential μ , evaluated using the mean-field solutions for the gap and the chemical potential, in the BEC and crossover regimes. The spin-wave contribution is shown for three representative temperatures, $T = 3.5, 7.0$, and 17.5 K, of the order of $T_{\text{BKT}}^{\text{RG}}$ for the densities considered in this work.

mean-field grand potential $\Omega_{MF} = TS_{MF}$ with respect to Δ_0 (Eq. (6)).

The density equation, which fixes the chemical potential, is obtained from the full grand potential $\Omega = \Omega_{MF} + \Omega_{SW}$. Since the spin-wave action S_{SW} in Eq. (28) is quadratic in the phase field, the functional integration over θ_{SW} can be performed exactly, yielding

$$\Omega_{SW} = T \sum_{\mathbf{q}} \ln \left(1 - e^{-\beta \omega(\mathbf{q})} \right), \quad (30)$$

with $\omega(\mathbf{q}) = c_s q$ the collective-mode spectrum. From $n = -\frac{1}{A} \partial \Omega / \partial \mu_s$ one obtains $n = n_{MF} + n_{SW}$, with the mean-field contribution,

$$n_{MF} = \frac{g}{A} \sum_{\mathbf{k}} \frac{1}{2} \left(1 - \frac{\epsilon_{\mathbf{k}} \tanh(E(\mathbf{k}, T)/2T)}{E(\mathbf{k}, T)} \right) \quad (31)$$

coming from $-\frac{1}{A} \partial \Omega_{MF} / \partial \mu_s$, and the spin-wave contribution,

$$n_{SW} = -\frac{1}{A} \frac{\partial \Omega_{SW}}{\partial \mu_s}. \quad (32)$$

Figure 7 shows that, when evaluated using the mean-field solutions for the gap and chemical potential, n_{SW} is several orders of magnitude smaller than n_{MF} over the temperature and density range relevant to the superfluid phase. Up to temperatures of order T_{BKT}^{RG} , spin-wave fluctuations, despite growing as the system approaches the low-density regime, provide only a negligible correction to the density equation. We can thus safely approximate $n \simeq n_{MF}$ throughout this work.